

Sushanth Buddhala

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Fairfax, Virginia

PROFILE

Data Analytics Engineering graduate student with hands-on project experience building RAG pipelines, agentic AI workflows, and machine learning models. Skilled in Python, SQL, Power BI, and the OpenAI API, with a focus on designing prompt engineering workflows, implementing data validation, and translating complex datasets into actionable insights.

PROJECT EXPERIENCE

Compass - Agentic Multi-Agent Retail Analytics System

Feb 2026 – May 2026

- Architected a 5-agent system (Planner, Data, Product, Supply Chain, Synthesizer) on a DuckDB warehouse with 13 tables and 25K orders, using Gemini API with function calling and a custom Pydantic orchestrator.
- Built a column synonym normalization layer and hybrid Flash/Pro fallback model strategy, achieving 100% completion on hard multi-agent analytical cases.
- Designed a 40-case eval harness with 4-dimensional scoring, reaching 95% completion and 95.2% correctness across 3 iterative bug-fix rounds.

SARRDI - Dam Infrastructure Risk Analytics

Jan 2026 – May 2026

- Built a 5-step geospatial ETL pipeline integrating 5 public datasets across 2,712 Virginia dams using Python, GeoPandas, and spatial joins.
- Designed a rule-based risk scoring model across 5 weighted dimensions (hazard, flood exposure, watershed risk, social vulnerability, data quality), flagging 170 critical dams and 74 missing Emergency Action Plans.
- Delivered a Power BI dashboard with interactive geospatial filtering and a prioritized unclassified dams registry, supporting evidence-based infrastructure decision-making for Virginia DCR.

Housing Migration Analytics Platform (SQL + RAG)

Nov 2025 – Jan 2026

- Designed and implemented a normalized SQLite analytics database containing 395K+ U.S. housing, migration, AHS, and CHAS records, leveraging Cursor AI to accelerate development and optimize indexed tables for analytical querying.
- Built a schema-aware, schema-sensitive RAG pipeline using OpenAI embeddings to generate safe, explainable SQL, preventing hallucinations, destructive queries, and invalid SQL.
- Improved query accuracy and reliability by enforcing schema validation and SQL guardrails, enabling non-technical users to safely explore data.

TECHNICAL SKILLS

Languages & Tools: Python, SQL, DAX, Microsoft Excel, Databricks, Snowflake

Data & ML: Data Analysis, Data modeling, Data Warehousing, Data visualization, Data validation, ETL Pipelines, Machine Learning, Unsupervised learning, Data pipelines, Product Analysis

Visualization & Cloud: Power BI, AWS

AI & LLM: Cursor.ai, Claude code, Multi-Agent Systems, Agent Orchestration, Large Language Models, LLM Observability, LLM Tools, Model Evaluation, RAG pipelines

EDUCATION

George Mason University

Virginia, USA

M.S. in Data Analytics Engineering

- Relevant Coursework:** Analytics & Big Data, Intro to Analytics & Modeling, Principles of Data Management & Mining, Visualization for Analytics, Big Data Essentials, Data Mining for Business Analytics, Interpretable Machine Learning, Data Analytics Project

Gokaraju Rangaraju Institute of Engineering and Technology

Telangana, India

B.S. in Computer Science Engineering

- Relevant Coursework:** Linear Algebra Differential Calculus, Programming for Problem Solving, Differential Equation and Vector Calculus, Database Management System, Python Programming, Artificial Intelligence, Data Analytics with Python, Machine Learning, Big Data Analytics, Cloud Computing, Data Warehousing and Data Mining

PROFESSIONAL CERTIFICATES

• IBM Data Analyst Professional Certificate by Coursera • AWS Academy Graduate - Cloud Foundation • Generative AI for Data Analytics by Coursera